

Tomás Inácio

Curriculum Vitae

Corroios, Lisbon

+351914200155

✉ tomas.s.inacio@gmail.com

Education

- 2022–2025 **M.Sc. in Mathematics (grade: 19/20)**, *Faculty of Sciences, University of Lisbon, Lisbon, Portugal*
Thesis (grade 20/20): *Analytical and Geometric Aspects of the Isomonodromy Problem for Linear and Non-linear Connections*
- 2019–2022 **B.Sc. in Physics with Minor in Mathematics (grade 18/20)**, *Faculty of Sciences, University of Lisbon, Lisbon, Portugal*

Research Scholarships

- 2024 **Inverting the Stokes map**, *Science and Technology Foundation and Group of Mathematical Physics of the University of Lisbon, Lisbon, Portugal*, Supervision: Dr. Giordano Cotti
- In this grant, we looked in detail at the work of T. Bridgeland and V. Toledano-Laredo's method of inverting the Stokes map. The aim is to provide all explicit computations in particular cases of interest. The two dimensional case is fully covered and is showcased in my Master's thesis.
- 2023 **Twistor Theory and Isomonodromic Deformations**, *Science and Technology Foundation and Group of Mathematical Physics of the University of Lisbon, Lisbon, Portugal*, Supervision: Dr. Giordano Cotti
- The grant served as an overview of applications of Twistor Theory to isomonodromic deformations. In particular, a possible application of L. Mason and N. M. J. Woodhouse's method to the ODE with a regular pole at the origin and an irregular singularity at infinity.
 - The grant also covered an introduction to T. Bridgeland's Joyce structures (the subsequent topic of my thesis).
- 2022–2023 **Twistor Theory and Isomonodromic Deformations**, *Science and Technology Foundation and Group of Mathematical Physics of the University of Lisbon, Lisbon, Portugal*, Supervision: Dr. Giordano Cotti
- The grant served as an introduction to isomonodromic deformations and Twistor Theory

Skills

Programming Python, LaTeX, C
Tools Mathematica
Languages English (IELTS: 7.5, equivalent to C1), Portuguese (fluent)

Conferences and Presentations

- 2023 **Winter Meeting in Lisbon - Mathematics & Applications**, Lisbon, Portugal –
Oral Presentation: *A Twistor Theory approach to isomonodromic deformations*

Events

- 2024 **A Matemática Inspira-me, Department of Mathematics FCUL**, Lisbon, Portugal– Oral Presentation: *Mais um caso particular*
- 2024 **Encontros de Álgebra, Department of Mathematics FCUL**, Lisbon, Portugal– Oral Presentation: *Point-free Topology*
- 2024 **EVM 2024 - Mathematics Summer Internships, Department of Mathematics FCUL**, Lisbon, Portugal– Preparation and supervision of the project: *Twistor Theory and the Airy Equation*

Workshop and Summer Schools

- 2024 Workshop: **Twistors in Geometry & Physics**, Isaac Newton Institute for Mathematical Sciences – Virtual Attendance
- 2024 Training School: **Contemporary trends in integrable systems**, University of Lisbon – Physical Attendance

Volunteer Work

- 2024 Volunteer Teaching Assistant at Faculty of Sciences, University of Lisbon - Assisted with grading coursework and assignments for the Master's course "Differentiable Manifolds."
- 2019 Volunteer Work at the Calouste Gulbenkian Planetarium, Lisbon - Researched and curated recent astronomy news and discoveries to support the development of new educational exhibits for public outreach.

Awards and Honors

- 2021 BsC Honors Diploma, *Faculty of Sciences, University of Lisbon*

References

- Prof. Carlos Florentino Full Professor and Researcher at the Faculty of Sciences and Coordinator of CMAFcIO, Faculty of Sciences at the University of Lisbon and CMAFcIO - Centro de Matemática, Aplicações Fundamentais e Investigação Operacional - **Email: caflorentino@fc.ul.pt**
- Dr. Giordano Cotti FCT Junior Researcher, GFMUL - Group of Mathematical Physics of the University of Lisbon - **Email: gcotti.uni@gmail.com**